

Trader Performance Analysis Based on Bitcoin Market Sentiment

Objective

The objective of this analysis is to investigate the relationship between Bitcoin market sentiment, measured using the Fear & Greed Index, and trader performance on the Hyperliquid platform. The goal is to identify patterns in profitability, trading behavior, and risk-taking under different market sentiment regimes.

Dataset Overview

Two datasets were analyzed:

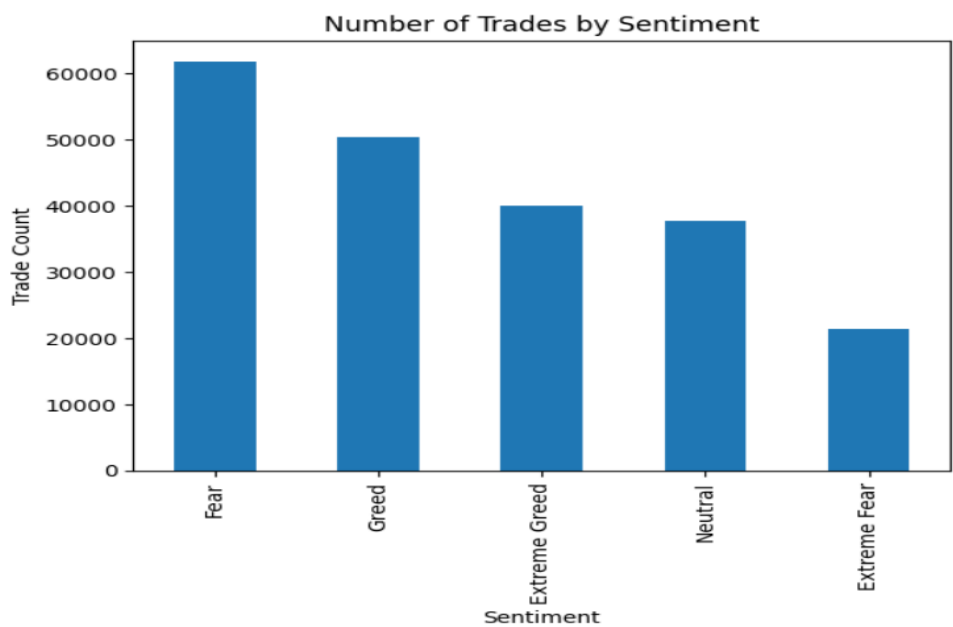
- 1. Bitcoin Fear & Greed Index dataset containing daily sentiment classifications.
- 2. Hyperliquid historical trading dataset containing 211,224 trades from 32 unique trader accounts.

The datasets were merged using trade date and sentiment date.

Key Findings

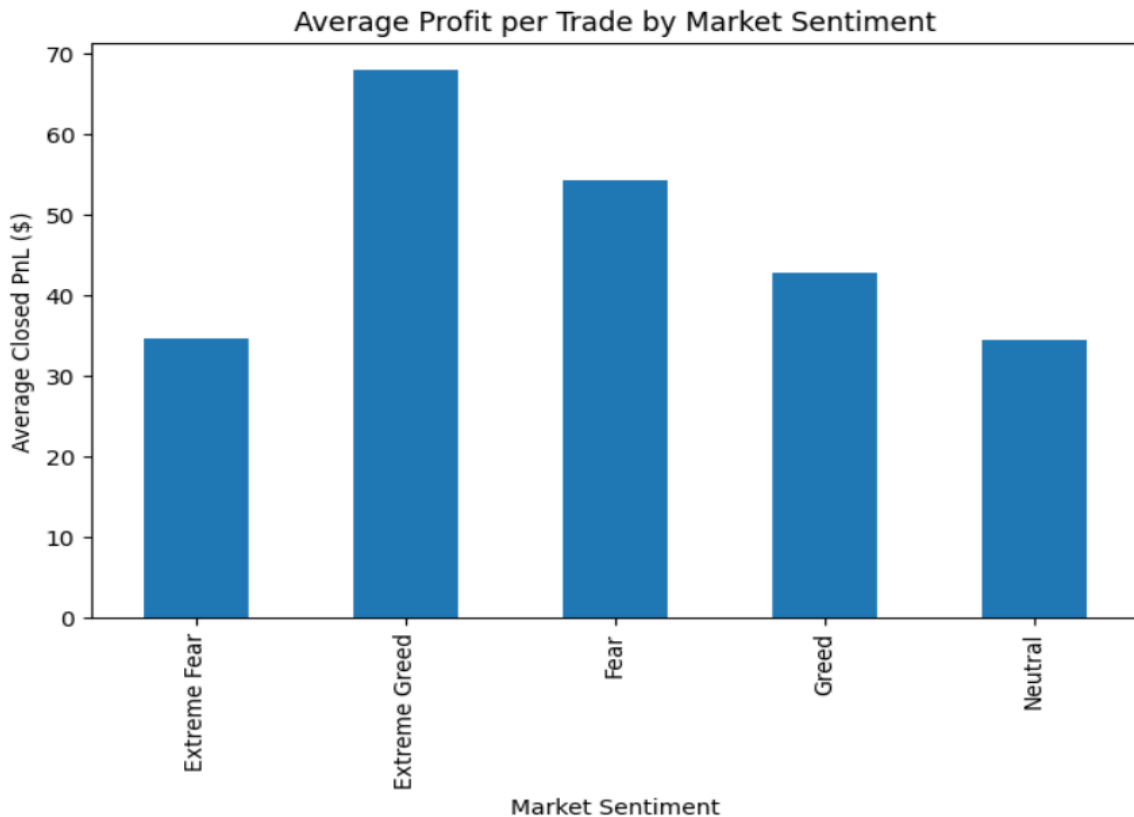
1. Trading Activity by Sentiment

Fear periods recorded the highest trading activity with 61,837 trades, while Extreme Fear periods recorded the lowest activity with 21,400 trades.



2. Profitability Analysis

Fear periods generated the highest cumulative profit of approximately \$3.36 million.



Extreme Greed generated the highest average profit per trade at approximately \$67.89.

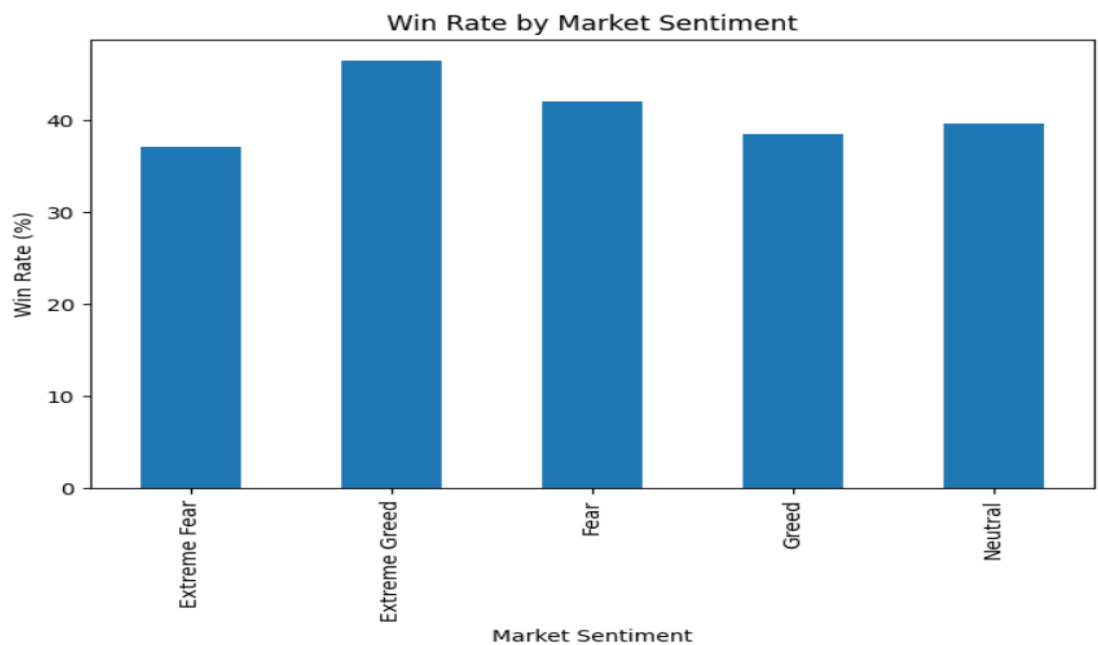
3. Win Rate Analysis

Extreme Greed recorded the highest win rate of **46.49%**, while Extreme Fear recorded the lowest win rate of **37.06%**.

This result suggests that traders were more successful during periods of strong market optimism. In Extreme Greed conditions, market trends are often well-established and momentum is stronger, making it easier for traders to identify profitable opportunities and follow prevailing price movements.

Conversely, Extreme Fear periods were associated with the lowest win rates. During such periods, markets tend to be highly volatile and uncertain, increasing the likelihood of false signals and unsuccessful trades. Traders attempting to enter positions during sharp market declines may face difficulties in accurately timing reversals.

The findings indicate that trend-following strategies tend to perform better during bullish sentiment regimes, whereas bearish sentiment regimes may require stricter risk management and more selective trade execution.



4. Position Sizing Behavior

Average trade size was highest during Fear periods at approximately \$7,816.

This indicates traders were willing to allocate larger amounts of capital during pessimistic market conditions, consistent with a contrarian accumulation strategy.

	mean	median
classification		
Extreme Fear	5349.731843	766.150
Extreme Greed	3112.251565	500.050
Fear	7816.109931	735.960
Greed	5736.884375	555.000
Neutral	4782.732661	547.655

5. Top Trader Performance

The most profitable trader generated approximately \$2.14 million in realized profits.

Several other traders generated profits exceeding \$800,000, indicating substantial performance differences among market participants.

Closed PnL	
Account	
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	2.143383e+06
0x083384f897ee0f19899168e3b1bec365f52a9012	1.600230e+06
0xbaaaf6571ab7d571043ff1e313a9609a10637864	9.401638e+05
0x513b8629fe877bb581bf244e326a047b249c4ff1	8.404226e+05
0xbec1707d6b44d4d52bfe19e41f8a828645437aab	8.360806e+05
0x4acb90e786d897ecffb614dc822eb231b4ffb9f4	6.777471e+05
0x72743ae2822edd658c0c50608fd7c5c501b2afbd	4.293556e+05
0x430f09841d65beb3f27765503d0f850b8bce7713	4.165419e+05
0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	3.790954e+05
0x72c6a4624e1dffa724e6d00d64ceae698af892a0	3.605395e+05

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Conclusions

The analysis reveals that market sentiment strongly influences trading behavior and performance.

Fear periods were associated with larger position sizes, higher trading activity, and the highest cumulative profits.

Extreme Greed periods delivered the highest average profitability and win rates, suggesting traders benefited from strong market momentum.

Overall, market sentiment is most effective when used as a market-regime indicator rather than as a standalone trading signal.

Recommendations

1. Increase exposure gradually during Fear periods while maintaining risk controls.
2. Use trend-following strategies during Extreme Greed conditions.
3. Apply stricter risk management during Extreme Fear periods.
4. Incorporate sentiment classification as a regime filter in systematic trading strategies.